

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EE) Examination****May 2018****EE309.01 Electric Power Utilization & Traction****Date: 08.05.2018, Tuesday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Answer the question below. [07]**
- a. Long distance railways use [01]
- (A) 200 V DC (C) 25 kV Two phase AC
- (B) 25 kV Single phase AC (D) 25 kV Three phase AC
- b. When the speed of the train is estimated taking into account the time of stop at a station in addition to the actual running time between stops, is known as [01]
- (A) Average speed (C) Notching speed
- (B) Schedule speed (D) Free running speed
- c. For 600 V dc line for tram cars [01]
- (A) track is connected to negative of the supply
- (B) track is connected to positive of the supply
- (C) track is connected to mid voltage of 300 V
- (D) None of the above
- d. The consideration involved in the selection of the type of electric drive for a particular application depends on [01]
- (A) Speed control range and its nature
- (B) Starting torque
- (C) Environmental conditions

(D) All of the above

- e. The coefficient of adhesion is better in _____ traction than that in steam traction. [01]
- f. During free run _____ remains constant. [01]
- g. The total weight of locomotive and train to be pulled by the locomotive is called _____. [01]

Q – 2.a Write advantages and disadvantages of electric braking over mechanical braking. [04]

Q – 2.b Answer any two questions. [10]

- (i) (A) The outer surface of 12 hp (8.82kw) total enclosed motor is equivalent to a cylinder of 65 cm diameter and 1m length. The motor weights 400 kg, the material having a specific heat of 700 J/kg/°C. The outer Surface is capable of dissipating heat at the rate of 12.5 W/m²/°C. Find the final temperature rise and the thermal time constant of the motor when operating at full load with an efficiency of 90%. [03]

(B) An electric motor has load variation as given below: torque 240 Nm for 20 minutes, 140 Nm for 10 minutes, 300 Nm for 10 minutes, 200Nm for 20 minutes. If the speed of the motor is 720 rpm, find the power rating of motor. [02]

- (ii) A train is required to run between two stations 1.6 km apart at an average speed of 40 kmph. The run is to be made to simplified quadrilateral speed time curve. If the maximum speed is to be limited to 64 kmph. Acceleration from rest at 2 kmphs. Coasting and braking retardation to 0.16 kmphs and 3.2 kmphs respectively. Determine the duration of acceleration, coasting and braking period. If stop time is 30 seconds find the schedule speed. [05]

- (iii) Explain counter current braking for D.C. series & shunt motor. [05]

Q - 3 Answer any two. [14]

- a. Derive Equation for Temperature rise of electric drive. Also draw Heating and cooling curve. [07]
- b. Explain transition methods. [07]
- c. Write features of an ideal traction system and Explain total tractive effort for propulsion of train. [07]

SECTION – II

- Q - 4 Answer the question below. [07]**
- a. **Visible light** falls in the _____ to _____ nm range. **[01]**
- (A) 400, 500 (C) 400,700
(B) 400,710 (D) 315,600
- b. Light is produced in electric discharge lamps by **[01]**
- (A) Heating effect of current (C) Ionization in a gas or vapour
(B) Magnetic effect of current (D) Carbon electrodes
- c. The transformer used for as welding sets is **[01]**
- (A) Booster type
(B) Step up transformer
(C) Step down transformer
(D) All of the above
- d. In dielectric heating, if the capacitor is loss-free, the heat produced will be **[01]**
- (A) Zero
(B) Infinity
(C) Proportional to the value of capacitance
(D) Proportional to the frequency
- e. For arc heating, the electrodes are made of **[01]**
- (A) Copper (C) Graphite
(B) Aluminum (D) ACSR conductor
- f. Which of the following will need the highest level of illumination? **[01]**
- (A) Proof reading (C) Hospital wards
(B) Bed rooms (D) Railway platforms.
- g. If d is the distance of a surface from a source, the illumination of the surface will vary as **[01]**
- (A) d (C) $1/d$
(B) $2d$ (D) $1/d^2$

Q – 5.a Explain dielectric heating with applications. [05]

OR

Q – 5.a Explain Two types of Butt Welding. [05]

Q – 5.b Draw the diagram of direct arc heating, coreless type and vertical core type induction Furness. [04]

Q – 5.c Explain seam welding and spot welding. [05]

OR

Q – 5.c Explain laws of illumination. [05]

Q – 6.a Explain photometry and also explain equality of brightness type of photometer head. [07]

Q – 6.b An electric train has quadrilateral speed time curve as follows; [05]

(i) Uniform acceleration from rest at 2 kmphs for 30 seconds; (ii) coasting for 50 seconds; (iii) uniform braking to rest for 20 seconds. If the train is moving a uniform down gradient of 1%, train resistance is 40 N/tonne, rotational inertia effect 10 % of dead weight and duration of stop 30 seconds, and overall efficiency of transmission gear and motor as 85% find the schedule speed and specific energy consumption of run.

Q – 6.c A suburban electric train has a maximum speed of 70 kmphs. The schedule speed including a station stop of 25 seconds is 45 kmph. if the acceleration is 1.5 kmphs, find the value of retardation when the average distance between stop is 3 km. [02]

OR

Q – 6.a The speed time curve of an electric train on uniform rising gradient of 1 in 100 comprises [07]
(i) Uniform acceleration from rest at 2 kmphs for 30 seconds; (ii) coasting with power off for 70 seconds; (iii) braking at 3 kmphs to stand still. The weight of the train is 250 tonne, the train resistance on level track being 5 kg/tonne, and allows 10% rotational inertia. Calculate the maximum power developed by traction motors and total distance travelled by the train assume transmission efficiency as 97%

Q – 6.b Explain basic types of reflection and type of lighting schemes. [05]

Q – 6.c Explain the terms used in electrolytic process. [02]

1) Current efficiency 2) Energy efficiency
